

# Similarity

Project: Helpful Classroom Sorter

## 1. Your mission

### Your learning target

I can calculate similarity using explicitly selected features and a stated scoring rule.

I can explain how a change in feature weights can change the highest scoring item.

I can handle ties and missing features without claiming that a similarity score proves suitability.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Recommendations often begin by comparing selected traits, not whole objects.



LEARNING TARGET A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

Learner name:

### Readiness for the connected explanation

A system gives an item two matching-feature points. Does that mean a two-thirds chance that the item is suitable? What must you read first?

### This week's end goal

By the end, I can calculate similarity under a declared feature rule and explain a changed ranking.

## 2. Notice the evidence

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



**MYSTERY** A book can be nearest to another by reading level, topic, length, or mood; the chosen features and scales determine the neighbor.

### What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

**GOOD THINKING** A prediction can change when stronger evidence appears.

### A closer explanation

Two objects can be alike in colour and different in shape. A computer comparison needs a precise answer to similar in what way. Select features, state how their values are compared, and define what a larger score means. The result belongs to that rule; it is not a universal judgment about the objects.

Our target card is red, round and smooth. Three candidate cards are A: red, round, rough; B: blue, square, smooth; C: red, square, rough. First give one point for each feature that exactly matches the target and zero for each known mismatch. Add the points. Colour, shape and texture are the only selected features.

This paper exercise uses complete category values with agreed spelling. Real records can be missing or uncertain. Do not silently treat an unknown texture as a known mismatch or copy a value from the target. First identify whether the information supports the comparison.

### 3. See how it works

#### VISUAL MECHANISM

**Input** → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



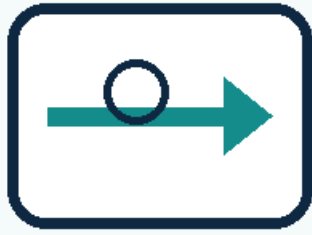
#### **INPUT**

represented traits for candidate items



#### **PROCESS**

compare distances using selected features



## OUTPUT

nearest or most similar items

INPUT represented traits for candidate items

PROCESS compare distances using selected features

OUTPUT nearest or most similar items

CORE IDEA Recommendations often begin by comparing selected traits, not whole objects.

UNDER THE HOOD Similarity is a designed measure over represented features. Distance changes with feature selection, scaling, weighting, and missing values. Nearest does not mean identical, best, or fair.

Draw a fresh example with all three stages:

### A closer explanation

#### Three features contribute to a score

**Candidate · Matches with target · One point per match**

**Candidate:** A: red round rough

**Matches with target:** Colour and shape

**One point per match:** 2 points

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**Candidate:** B: blue square smooth

**Matches with target:** Texture only

**One point per match:** 1 point

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**Candidate:** C: red square rough

**Matches with target:** Colour only

**One point per match:** 1 point

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The target is red, round and smooth. A has the largest score under equal weights. These points count selected feature matches; they are not probabilities or evidence that A suits every purpose.

A receives 1 for colour, 1 for shape and 0 for texture: total 2. B receives 0, 0 and 1: total 1. C receives 1, 0 and 0: total 1. Record the contributions, not just the totals, so another learner can check the computation.

A is most similar among these three candidates under this rule. That sentence names the candidate set and the rule. It does not say A is identical to the target, has the right texture or will be the best choice for a task that needs smooth material.

### Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

PICTURE GALLERY

### Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



**Same topic, different level**



**Same length, different  
mood**



**Weighted topic**



**Missing feature**

CASE 1 — Same topic, different level Clue: close on topic. Meaning: far on reading level.

CASE 2 — Same length, different mood Clue: close on length. Meaning: far on mood.

CASE 3 — Weighted topic Clue: topic counts more. Meaning: neighbor changes.

CASE 4 — Missing feature Clue: distance incomplete. Meaning: need policy.

BOUNDARY Similarity is a tool designed for a purpose, not a universal property.

Which two pictures are easiest to confuse? What evidence separates them?



## Change one thing in the pictured case

Change: Keep original weights but change the target texture to rough.

Prediction: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

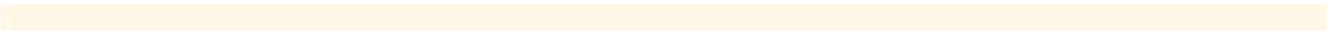
## 5. Words that unlock the idea

### VOCABULARY

#### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





**D**

**distance**

**W**

**weight**



### Word · Meaning · My example

**Word:** similarity

**Meaning:** degree of likeness under chosen traits

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** distance

**Meaning:** numeric or ordered difference between representations

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** weight

**Meaning:** importance assigned to a feature

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** nearest neighbor

**Meaning:** item with smallest chosen distance

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Misconception detective**

CLAIM TO REPAIR The nearest item is automatically the best recommendation.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

## Words used in the closer explanation

### Word · Meaning

**Word:** Feature match

**Meaning:** Agreement between two recorded values for a selected feature

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**Word:** Tie

**Meaning:** Two or more candidates sharing the same score

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**Word:** Candidate

**Meaning:** An item being considered in a comparison

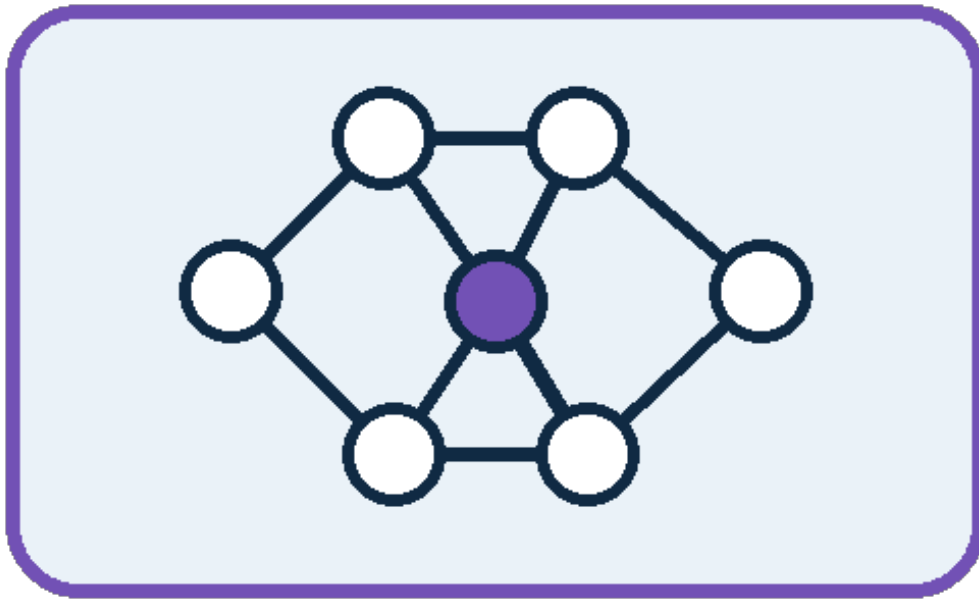
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## 6. Follow a worked example

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A book can be nearest to another by reading level, topic, length, or mood; the chosen features and scales determine the neighbor.

STEP 1 — OBSERVE represented traits for candidate items

STEP 2 — TRACE compare distances using selected features

STEP 3 — CONCLUDE nearest or most similar items

MODEL REASONING Plot items consistently, define how each axis is measured, and compare distances. A different axis or weight can change the nearest item without any item moving in the real world.

DO NOT OVERCLAIM Similarity is a tool designed for a purpose, not a universal property.

What new evidence could change this answer?

### A closer explanation

**Candidate · Colour and shape points · Texture worth 3 points · New total**

**Candidate:** A

**Colour and shape points:** 1 + 1

**Texture worth 3 points:** 0

**New total:** 2

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**Candidate: B**

**Colour and shape points: 0 + 0**

**Texture worth 3 points: 3**

**New total: 3**

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**Candidate: C**

**Colour and shape points: 1 + 0**

**Texture worth 3 points: 0**

**New total: 1**

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Suppose texture matters more for the stated comparison. Keep colour and shape worth one point each, but give three points for a texture match. B now scores 3 and becomes the unique highest scoring candidate. No card changed; the rule changed. Record both versions rather than overwriting the first result.

Weights express priorities in this designed rule. They need a reason connected to the task and should be set before inspecting the final evaluation results. A convenient winner does not justify a weight. If smooth texture is required without exception, use a requirement check rather than hoping a weighted score will always enforce it.

## **Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## **A pictorial worked case: Similarity**

A fictional craft-card matcher compares each candidate with a target.

### **The facts we will use**

Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact feature match scores 1; return all highest scorers.

For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

## Declared weights change similarity

|   | All weights 1 | Texture weight 3 |
|---|---------------|------------------|
| A | 2             | 4                |
| B | 2             | 4                |
| C | 2             | 2                |

Read a cell using both its row and column labels.

## Similarity

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Target features

Compare colour, shape and texture separately.



#### 2. Equal weights

A matches colour+texture=2; B shape+texture=2; C colour+shape=2.



#### 3. Initial result

All three tie at 2.



#### 4. Texture weight 3

New scores A=4, B=4, C=2; A and B remain tied.

### CHANGE ONE THING

Keep original weights but change the target texture to rough.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Target features

**What happens and why:** Compare colour, shape and texture separately.

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**Step:** 2. Equal weights

**What happens and why:** A matches colour+texture=2; B shape+texture=2; C colour+shape=2.

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**Step: 3. Initial result**

**What happens and why:** All three tie at 2.

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**Step: 4. Texture weight 3**

**What happens and why:** New scores A=4, B=4, C=2; A and B remain tied.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Similarity is defined by the chosen representation and weights. The cards did not change when texture gained weight; the comparison rule did. Two high scores need not mean two identical cards. A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

## What this example does not establish

A high feature-match score does not establish usefulness or suitability beyond the features represented.

Read the labelled picture: Point to the matches supporting C's new score and explain why there is no tie.

Change one thing: Keep original weights but change the target texture to rough. Predict the result before checking the explanation. What remains the same?

# 7. Find and repair a mistake

COMPARE

## Same Result, Different Reason

A comparison helps reveal the true concept boundary.

### Case · Evidence · What it shows

**Case:** Same topic, different level

**Evidence:** close on topic

**What it shows:** far on reading level

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**Case:** Same length, different mood

**Evidence:** close on length

**What it shows:** far on mood

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**Case:** Weighted topic

**Evidence:** topic counts more

**What it shows:** neighbor changes

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**Case:** Missing feature

**Evidence:** distance incomplete

**What it shows:** need policy

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## Repair two tempting claims

CLAIM A The nearest item is automatically the best recommendation.

Repair:

CLAIM B Two items have one permanent similarity score.

Repair:

STRONG ANSWER FRAME I claim \_\_\_\_\_ because the evidence shows \_\_\_\_\_. The process is \_\_\_\_\_. One limit is \_\_\_\_\_.

## A closer explanation

Feature selection leaves information out. A card comparison ignores size and material unless those are included. A book comparison using topic and format may ignore language or reading difficulty. More features are not automatically better: irrelevant or repeatedly copied features can change a score without improving the decision.

A useful report states the purpose, candidates, selected features, weights, missing-value policy and tie policy. Check the proposed method on relevant cases that include ties and imperfect records. If it supports a recommendation, keep the explanation limited to what was actually compared.

**A learner says A is objectively the closest card because it won the original comparison. Use the changed texture weight and the added candidate D to explain why the claim needs qualification**

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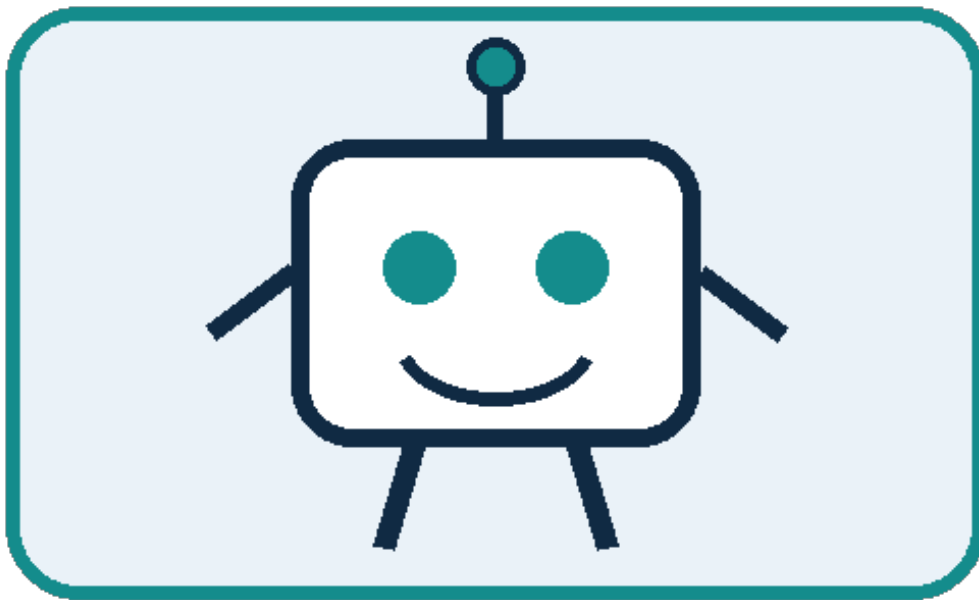
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## 8. Practise together

### GUIDED PRACTICE

#### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Recommendations often begin by comparing selected traits, not whole objects.

Use a fresh example to explain similarity.

Trace a new input → process → output example.

Find and repair the error in: Two items have one permanent similarity score.

Invent one edge case and name the safe response.

**PARTNER CHECK** Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

### A closer explanation

Add candidate D: blue, round, smooth. Under the original one-point rule, D scores 2, tying A. Our policy is to return every candidate with the largest score, so the result is A and D. Returning only the first row would silently introduce a different policy.

Using A, B, C and D, calculate the four totals under the original rule and then under texture worth 3. Return every highest scoring candidate each time. Explain which matches caused the change and why neither result establishes a probability of suitability

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If a required comparison value is missing, our classroom policy pauses the final ranking and requests the missing value. Retain the known contributions and the reason for the pause. A different policy could be designed, but it would need to be explicit and checked for its effects. Unknown and different are separate states.

### Practise with less help: Explain a changed winner

Use workbook W2 for the connected example “Similarity and selected features”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Similarity and selected features. Proposed explanation: “Similar has one objective meaning for every task”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Point to the matches supporting C's new score and explain why there is no tie.

**My explanation using a pictured step and the given facts**

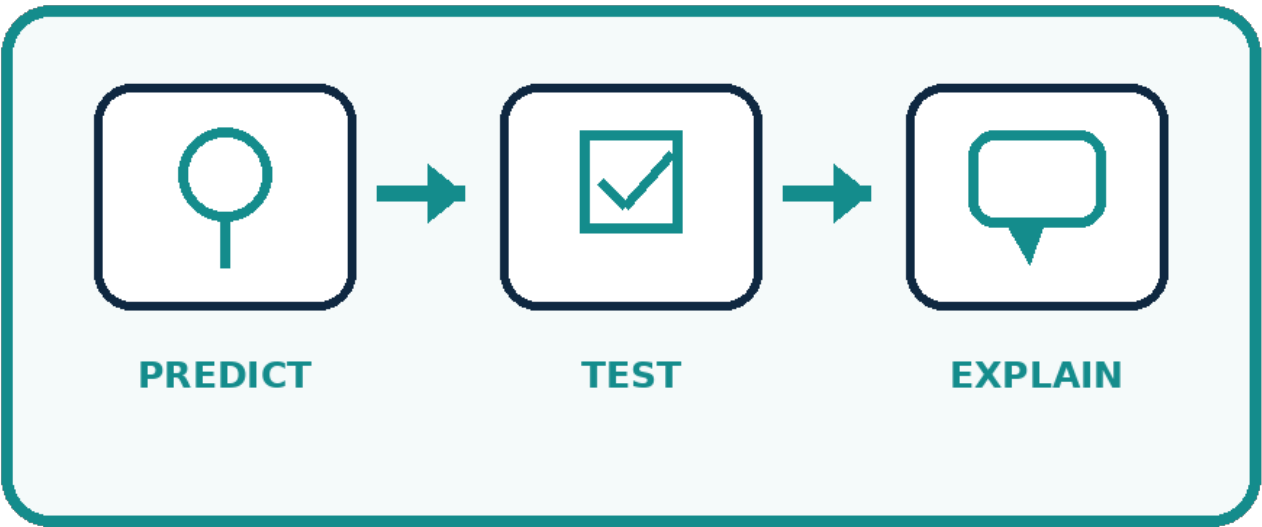
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Trait Map Neighbors

Predict first, test honestly, and explain what the evidence means.



- State your prediction before the result is known.
- Complete the task: Create a trait map and find two different 'nearest' items.
- Keep a normal case and an edge case clearly separated.
- Record what happened—even when it does not match your prediction.
- Explain the result, one limitation, and the next useful test.

**Case · Prediction · Observed result · What it means**

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Edge

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

\_\_\_\_\_

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

**Plan → try → check**

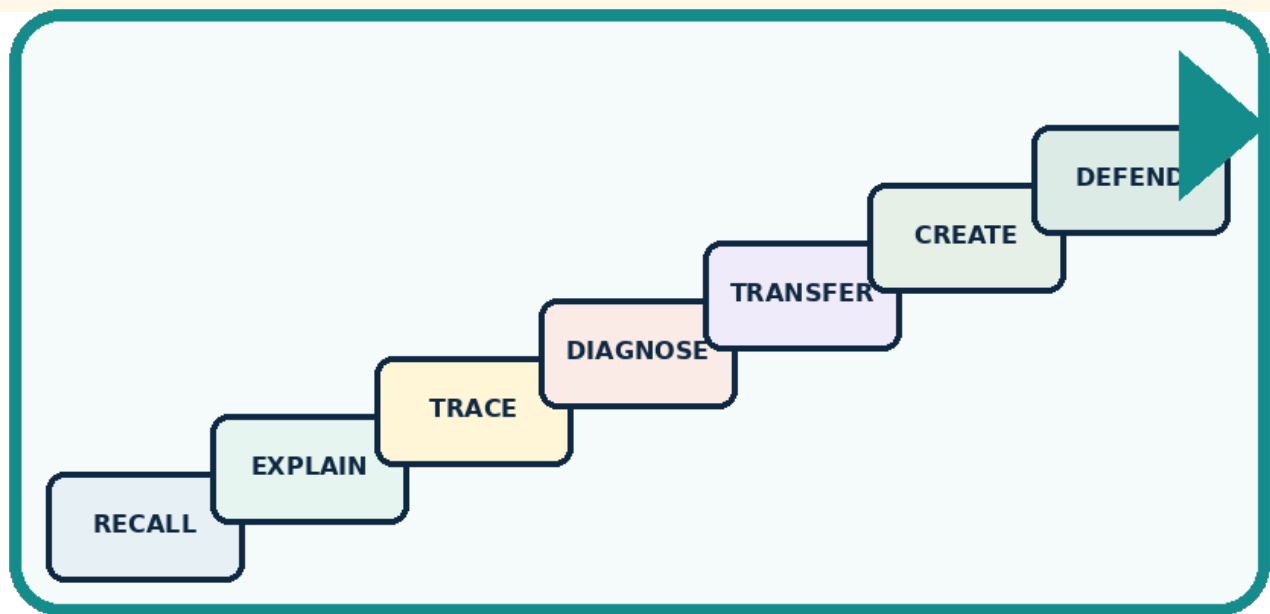
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

PRACTICE LADDER

### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define similarity.

EXPLAIN Recommendations often begin by comparing selected traits, not whole objects.

TRACE Show the input, process, and output.

DIAGNOSE Repair: The nearest item is automatically the best recommendation.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

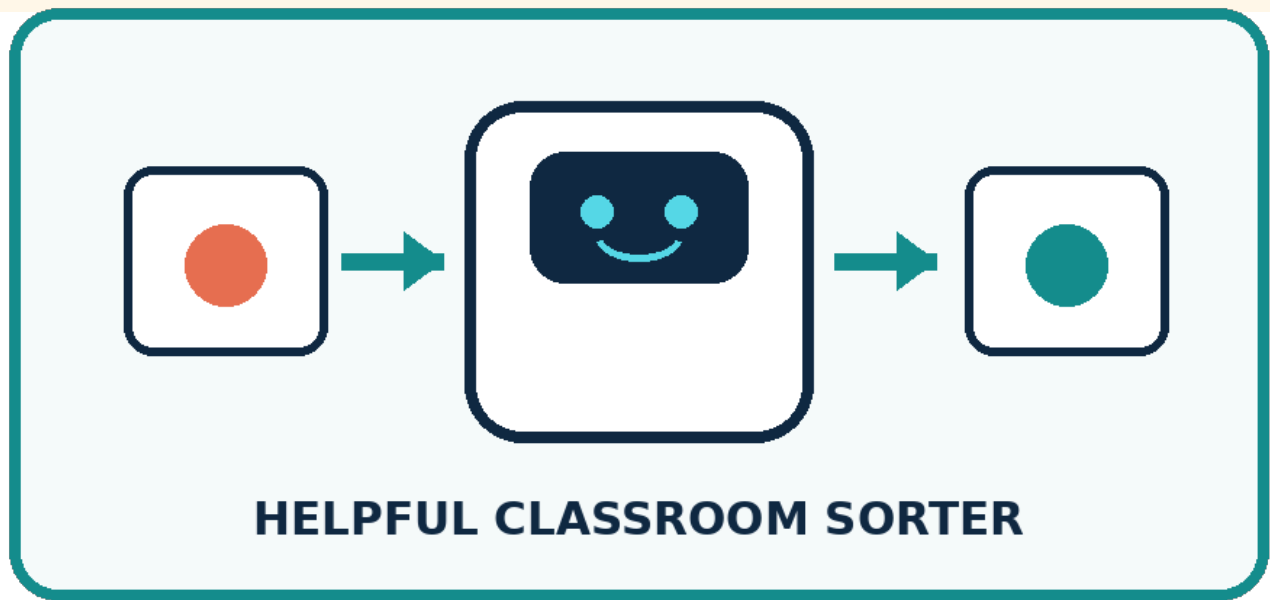
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

## 11. Connect your project

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



**THIS WEEK'S CHECKPOINT** Choose two useful item traits for similarity-based sorting and explain why each deserves its weight.

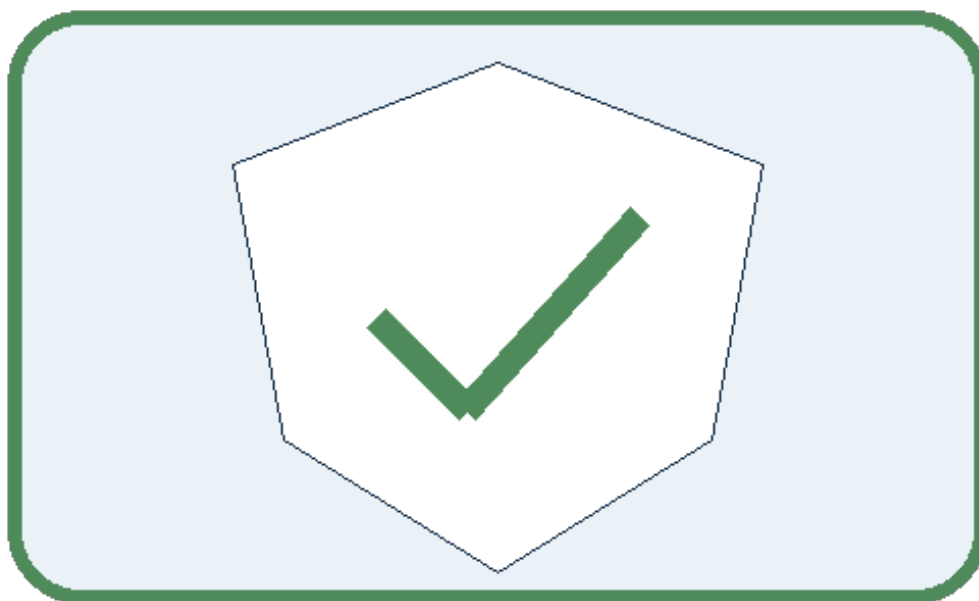
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?





RESPONSIBLE DESIGN Similarity is a tool designed for a purpose, not a universal property.

My project decision:

### Evidence for your project

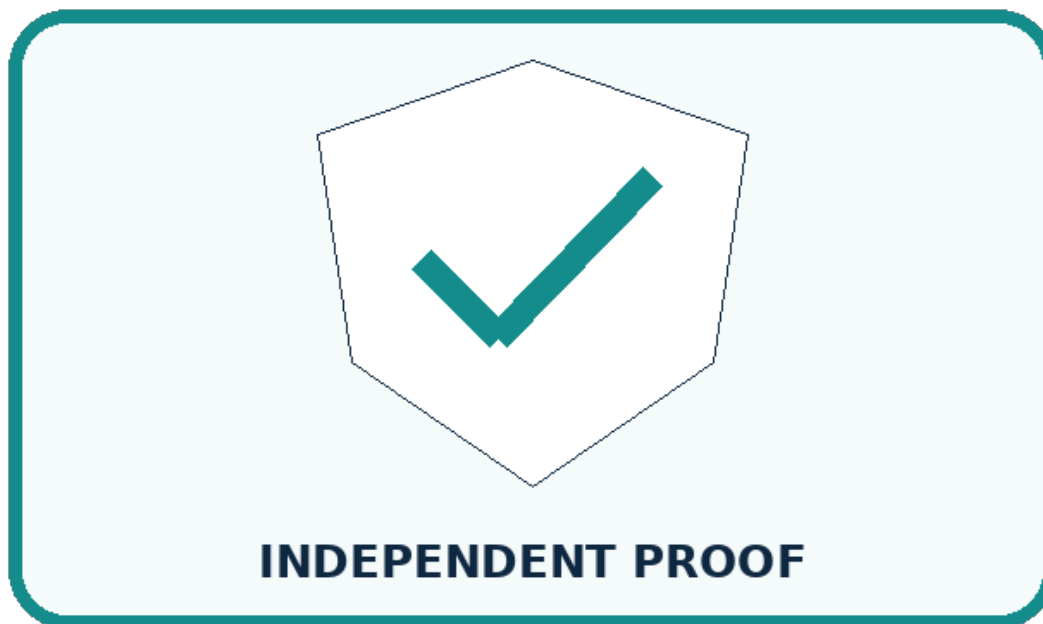
Declare sorter similarity features, weights and tie behavior before using scores to explain recommendations.

## 12. Show what you know

MASTERY

### Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



**REQUIRED PROOF** Create a two-trait map and show how changing one trait weight produces a different nearest item.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Plot items consistently, define how each axis is measured, and compare distances. A different axis or weight can change the nearest item without any item moving in the real world.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Recommendations.

## 13. Study a complete case

### Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

#### The question

Rank by number of matching traits.

#### Worked explanation

A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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#### Trace the supplied case visually

#### Count matching book traits

## Evidence or stage · Value or result

**Evidence or stage:** Book A

**Value or result:** 2

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**Evidence or stage:** Book B

**Value or result:** 1

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**Evidence or stage:** Book C

**Value or result:** 0

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Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

Rank by number of matching traits.

A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

## 14. Try a new case independently

### Independent case: Transfer the comparison to books

A target book has topic space, format comic and language English. Candidates are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers. Then make topic worth 3 points while keeping the other weights at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case

already practised with feedback cannot establish fresh transfer.

**Week 19 Independent application**

Close the worked explanation. Apply the idea to this different question without copying.

Compare books using topic only. Book A is about space with a red cover; B is about space with a blue cover; C is about dogs with a red cover. Which matches A best?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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**Create and test**

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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**Connect to your project**

Choose two useful item traits for similarity-based sorting and explain why each deserves its weight.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

### Transfer practice from the original programme

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional craft-card matcher compares each candidate with a target.

**By the end, I can · How I will show it**

**By the end, I can:** Calculate similarity using explicitly selected features and a stated scoring rule

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Explain how a change in feature weights can change the highest scoring item

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Handle ties and missing features without claiming that a similarity score proves suitability

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Calculate similarity using explicitly selected features and a stated scoring rule

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft-card matcher compares each candidate with a target.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? New scores  $A=4$ ,  $B=4$ ,  $C=2$ ; A and B remain tied.

1. Target features
2. Initial result
3. Texture weight 3
4. Equal weights

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Texture weight 3 New scores A=4, B=4, C=2; A and B remain tied. Similarity is defined by the chosen representation and weights. The cards did not change when texture gained weight; the comparison rule did. Two high scores need not mean two identical cards. A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

## Practice 2: explain the deciding evidence

Changed-case practice: Keep original weights but change the target texture to rough. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. C scores 3; A and B each score 1.

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must

say what follows from those facts.

After your attempt, compare with the supported explanation: C scores 3; A and B each score 1. C now matches all three target features; the others lose their smooth-texture match.

## **Repair a missing building block**

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from evidence that a prediction was correct. Short practice: Was the unknown label guaranteed correct because its confidence was 80%?. Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a prediction. Short practice: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?. Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing predictions and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?. Return to this week and explain what you changed.

## **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.